1. A motion program, program `p1a.py`:

Let the robot execute a velocity program for 30 s with at least 3 different velocity commands (including linear and angular velocities) without human interaction. Plot the commands, the IMU data (accelerations and angular velocities in the sensor frame) and the odometry data and interpret it.

2. Inertial navigation, program p1b.py:

Play with the noise components of the IMU and set the update rate to 100 Hz (the quantization effects will be strong enough to introduce some noise, file `rsim/rsim_description/urdf/imu/imu.gazebo.xacro`, the imu is integrated into the model in `rsim_bringup/urdf/robert.urdf.xacro`). Perform inertial navigation: Transform the IMU data (accelerations and angular velocities) into an inertial frame and integrate them using the quaternion differential equation. *A template for the program is available in Moodle*. Let the robot move as in `p1a` and compare the result against the odometry data and plot both 3d-position and orientation estimates.

Remark:

- Do not be disappointed if the results of the inertial navigation are poor. The simulation introduces a lot of noise.
- The template contains formulas to obtain the rotation matrix (DCM) and differential equation from the quaternion (quat2DCM and quatDEQ). Double-check the orientation of the rotation, it might be necessary to consider the transpose.

3. Lidar sensors in XACRO:

Investigate the (combined) lidar data in `rviz2`. Change the position of the Lidar sensor(s) so that the sofa table can be seen clearly. Visualize the measurements for both Lidar positions.

4. Robert as vacuum cleaner, program p1c.py:

Drive around the living room randomly without colliding with objects making sure that you will eventually cover all reachable parts. *Hint*: One strategy could be to move straight on until the way is blocked and then turn in a new, random direction.

5. Coordinate transforms

A robot at world coordinates (5, 3, 0) is rotated by $\pi/6$ around the z-axis. What are the robot coordinates of an object with world coordinates (9, 1, 1)? What are the world coordinates of an object with robot coordinates (1, 0, 0)? (World and robot system are defined with z-axis up. Use homogeneous coordinates and transformation matrices to convert the coordinates. Give the result as inhomogeneous vector with reals rounded to 3 valid digits.)

6. Differentíal/Integrating sensors

An accelerometer is subject to noise. We integrate the data to obtain velocities. The noise is such that after one second the integrated data is normal distributed with expectation $0m/s$ and variance $5m^2/s^2$. What is the error distribution after 5 sec. (Define the probability distribution exactly, do not only give the variance. Do not forget the units.)

Write a report in pdf format which contains the source code, the changed parts of the configuration files and the requested visualizations for the practical part. Every program, visualization and answer of the questions shall be explained ***shortly*** (one paragraph per program/visualization/answer). Add the numbers of the assignments as chapter numbers.

The first page of the report shall contain the name of the lecture ("Autonome Systeme und mobile Roboter"), the name of the project (P1 etc.), your name(s), the date of submission and a remark that the project was prepared using authorized resources only. If the project was processed by more than one person, add the authorship for every chapter. The report shall be named ***report_***<*your_name(s)*>***.pdf***. <your_name(s)> is your surname or the surnames of the two people working together, so a valid report name is “report_mueller_meier.pdf”. Zip the programs and the report (name: ***P1***_<*your_name(s)*>***.zip***) and follow the instructions in Moodle to submit the work.